A uniform transverse limit of the dS_3 crossing kernel and its local action on test functions

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Abstract

Combining the two removable Gamma poles before estimation gives a bound uniform through the collision surface. The physical scalar kernel has a phase-shifted sine approximation with uniform $O(M^{-1})$ error on a fixed transverse interval. It converges to a unit delta distribution there with respect to Lebesgue transverse measure. This local statement does not yet control the physical mass measures, creation amplitudes or noncompact spectral tails of a Bose observable.

1 Scope and the local theorem

We retain $\ell = 1$, $G_N = 0$, proper $SO^+(3,1)$, BD scalar observers, KG matter, a fixed positive observer-label lower bound, and half-period regulated creation blocks. The scalar matrix element U_{00} keeps its physical CG-period phase; it is not a separately unitary scalar transform. The label M has static energy $\sqrt{1+M^2}$.

Use $c = M$, $b = M + u$, $a = M + u + v$, $d = M + v + r$, $p, q > 0$, and

$$A_j = \tfrac{1}{2} + i\alpha_j, \qquad \alpha = \tfrac{1}{2}(v+q, p-u, -u-p, v-q), \qquad z = -ir/2.$$

Choose $R > 0$ with $\max_j |\alpha_j| + R/2 \le 1/2$. All assertions are uniform for $|r| \le R$ and compact transfer families satisfying these conditions, once M is sufficiently large. Let $L = \log M$ and

$$\delta(r) = \Im\left[2\log\Gamma(1 - ir/2) + \sum_j \{\log\Gamma(A_j + ir/2) - \log\Gamma(A_j)\}\right],$$

continuously from $\delta(0) = 0$. Define

$$K_M(r) = \frac{\sin(Lr - \delta(r))}{\pi r}, \quad r \neq 0, \qquad K_M(0) = \frac{L - \delta'(0)}{\pi}. \tag{1}$$

We prove the model statement

$$\sup_{|r| \le R} |U_{00}(M, r) - K_M(r)| = O(M^{-1}). \tag{2}$$

Consequently, for $f \in C_c^2(\mathbb{R})$ supported in $[-R, R]$,

$$\int_{-R}^{R} f(r) U_{00}(M, r)\, dr = f(0) + O_f(1/\log M) + O(\|f\|_1 / M). \tag{3}$$

The coefficient one is with respect to dr. No mass Jacobian or physical spectral weight has been incorporated into (3).

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2 Merging before bounding

The preceding LPRS/Wilson decomposition [1, 2] is

$$X = c_A F + c_B G, \quad Y = y_A F + y_B H, \quad U_{00} = 2\Re(C_+ XY).$$

Set $\mathfrak{a} = zy_A$, $\mathfrak{b} = -zy_B$ and $d_0 = M + v$. The Gamma poles disappear:

$$\mathfrak{a} = \Gamma(1+z)\frac{\Gamma(1-ib)}{\Gamma(1-ib-z)}\frac{\Gamma(A_3-z)\Gamma(A_4-z)}{\Gamma(A_3)\Gamma(A_4)},$$
$$\mathfrak{b} = \Gamma(1-z)\frac{\Gamma(1+i(d_0+r))}{\Gamma(1+id_0-z)}\frac{\Gamma(A_1)\Gamma(A_2)}{\Gamma(A_1-z)\Gamma(A_2-z)}.$$

Both equal one at $r = 0$. Also $F(0) = H(0)$, and

$$Y = \frac{\mathfrak{a}F - \mathfrak{b}H}{z}, \qquad S = \frac{\mathfrak{a} - \mathfrak{b}}{z}. \tag{4}$$

Both expressions take their continuous derivative values at zero.

Here are explicit real constants, not sampled maxima. Put $m = M - \max(|u|, |v|) - R > 0$, $B = \max(|u|, |v|) + R$, and

$$C_0 = \cosh(\pi/2)\sqrt{1 + R/(2m)}\, e^{\pi R/4}(1 - e^{-2\pi m})^{-1/2},$$
$$P_1(t) = \gamma + 1 + \operatorname{arsinh} t, \qquad P_{1/2} = \gamma + 2\log 2 + 2 + \operatorname{arsinh} 1,$$
$$Q = P_1(R/2) + 2P_1(M+B) + 2P_{1/2}.$$

Gamma moduli give $|\mathfrak{a}|, |\mathfrak{b}| \le C_0$. The digamma difference series [3], compared with its decreasing integral, gives for $\sigma > 0$

$$|\psi(\sigma + it)| \le |\psi(\sigma)| + \sigma^{-1} + \operatorname{arsinh}(|t|/\sigma).$$

For example the extra sum is bounded by its first term plus $\int_\sigma^\infty |t|/[x\sqrt{x^2+t^2}]\, dx$. Differentiating the displayed regular coefficients therefore gives $|\mathfrak{a}'| + |\mathfrak{b}'| \le C_0 Q$ and $|S| \le 2C_0 Q$.

Write $F = \sum f_n$, $H = \sum h_n$. On the whole interval,

$$|f_n|, |h_n| \le \frac{1}{(n+1)^2}\prod_{j=1}^{n}\frac{j^2}{j^2+m^2}, \qquad |f_n'| \le 6H_n|f_n|, \quad |h_n'| \le H_n|h_n|.$$

Here $H_n = \sum_{j=1}^n j^{-1}$. The derivative bounds follow by logarithmic differentiation of the Pochhammer products; the numerator real parts are $1/2$ and denominator real parts are one. The sequences $H_n/(n+1)^2$ are summable, justifying termwise differentiation. Since $\mathfrak{a}(f_n) - \mathfrak{b}(h_n)$ vanishes at zero, the fundamental theorem of calculus removes the apparent $1/r$. Using $\sum_{n\ge1} H_n/(n+1)^2 \le 2$ gives

$$|Y - S| \le R_Y = \frac{2C_0\{(\zeta(2)-1)Q + 14\}}{1+m^2} = O(\log M/M^2). \tag{5}$$

For the merged sum through $n = N-1$, its exact-arithmetic tail is

$$T_Y(N) = \frac{2C_0}{N}\prod_{j=1}^{N}\frac{j^2}{j^2+m^2}\{Q + 7(2+\log N)\}. \tag{6}$$

3 Complete-kernel and coefficient errors

Let $D_0 = C_+ c_A$, $E_0 = C_+ c_B$, $R_F = (\zeta(2)-1)/(1+m^2)$, $Y_* = 2C_0 Q + R_Y$, and

$$D_* = \frac{C_0}{4\pi}\sqrt{\frac{\sinh(\pi R/2)}{\pi R/2}}.$$

The exact Gamma modulus formula implies $|D_0| \le D_*$. The preceding whole-series bound $|G| \le B_G$ uses a sinh product and a monic degree-ten ratio polynomial; it is uniform on the present compact window, with $|E_0|B_G = O(M^8 e^{-\pi M/2})$. We obtain the finite-$M$ estimate

$$|U_{00} - 2\Re(D_0 S)| \le 2D_*\{R_Y + R_F Y_*\} + 2|E_0|B_G Y_*. \tag{7}$$

Thus the complete series error is uniformly $O(\log M/M^2)$ plus an exponential term, including $r = 0$. Bound (7) is for the Gamma-exact approximation, not directly for (1).

It remains to make the Gamma expansion uniform. Define

$$\widetilde{D} = D_0 M^{-z}, \quad \widetilde{a} = \mathfrak{a} M^{-z}, \quad \widetilde{b} = \mathfrak{b} M^{z},$$

$$a_\infty = \Gamma(1+z)e^{-i\pi z/2}\frac{\Gamma(A_3 - z)\Gamma(A_4 - z)}{\Gamma(A_3)\Gamma(A_4)},$$

$$b_\infty = \Gamma(1-z)e^{-i\pi z/2}\frac{\Gamma(A_1)\Gamma(A_2)}{\Gamma(A_1 - z)\Gamma(A_2 - z)}.$$

The previous physical phase calculation gives D_∞ and

$$\widetilde{D} = D_\infty\left[1 + \frac{r}{4M} - \frac{ir(3r+4v)}{8M} + O_{C^1}(M^{-2})\right].$$

Its modulus follows directly from

$$\left|\frac{\widetilde{D}}{D_\infty}\right|^2 = \frac{b}{b - r/2}\frac{1 - e^{-2\pi b}}{1 - e^{-2\pi(b - r/2)}}.$$

For the phase and the two other coefficients, shifted Stirling and the digamma expansion [4] give, uniformly in real r,

$$\|\widetilde{D} - D_\infty\|_{C^1} + \|\widetilde{a} - a_\infty\|_{C^1} + \|\widetilde{b} - b_\infty\|_{C^1} = O(M^{-1}).$$

The derivative estimate uses the digamma remainder on the same sector; it does not follow merely by formally differentiating an unspecified pointwise big-O term. All small Gamma arguments stay away from poles, and large arguments stay away from the negative real cut.

Put $E_A = \widetilde{D}\widetilde{a} - D_\infty a_\infty$ and $E_B = \widetilde{D}\widetilde{b} - D_\infty b_\infty$. At zero both equal

$$\epsilon_0 = D_0(M, 0) - (4\pi)^{-1} = O(M^{-2}),$$

by the preceding peak-phase identity, while $|E_A(r) - \epsilon_0| + |E_B(r) - \epsilon_0| = O(|r|/M)$. Since $|e^{-irL} - 1| \le |r|L$,

$$\left|\frac{E_A(r)e^{-irL} - E_B(r)}{z}\right| = O(M^{-1} + L/M^2)$$

uniformly through zero. The limiting coefficients reproduce K_M by the previously proved phase and purely imaginary nonoscillating branch. Together with (7), this proves (2).

4 Local distributional action and its precise boundary

We give the short Fourier argument to fix the limiting normalization. For real or complex $f \in C_c^2(\mathbb{R})$ supported in $[-R, R]$, set

$$g(r) = f(r)\cos\delta(r), \qquad q(r) = \frac{g(r) - g(0)}{r}, \qquad h(r) = f(r)\frac{\sin\delta(r)}{r}.$$

The removable values at zero make q, h continuously differentiable; δ is smooth. Then

$$\int_{-R}^{R} fK_M\, dr = \frac{2f(0)}{\pi}\,\mathrm{Si}(LR) + \frac{1}{\pi}\int_{-R}^{R}\{q(r)\sin Lr - h(r)\cos Lr\}\, dr.$$

Integration by parts and the Dirichlet integral give

$$\left|\int fK_M\, dr - f(0)\right| \le \frac{6|f(0)|/R + \|q'\|_1 + \|h'\|_1}{\pi L}. \tag{8}$$

For example $|\int_x^\infty \sin t/t\, dt| \le 2/x$ supplies the sine-integral tail; the endpoint values of q contribute $2|f(0)|/R$, and h vanishes at both endpoints. This is the same standard removable-pole/Dirichlet mechanism used in the Wilson analysis [2]; the physical kernel and its uniform error have been controlled separately here.

Adding (2) proves (3). The constant in the $O(1/M)$ kernel estimate depends on the compact transfer family. We have not produced an optimized numerical value for that constant. The explicit estimates (5)–(7) remain available for the series part, with no arithmetic-rounding certification.

There is also a useful conditional extension. If f_M are all supported in this fixed interval, their C^2 norms are uniformly bounded, and $f_M(0) \to f_*(0)$, then

$$\int f_M(r) U_{00}(M, r)\, dr \longrightarrow f_*(0).$$

Indeed the quantities in (8) and $\|f_M\|_1$ are then uniformly bounded. This states what would suffice for one local piece of a physical wave packet; it does not assert that the actual creation amplitudes satisfy these hypotheses.

The conclusions do not include integration over the other transfer variables, the observer spectral Jacobians, a moving lower bound, noncompact mass tails, or growing-time factors. The full Bose free-observer limit therefore remains open. In particular a sharp moving cutoff or a test family with uncontrolled derivatives cannot be substituted into this theorem without further estimates. Full Fock domains, finite-G_N dynamics and a microscopic de Sitter holographic dictionary remain later objectives.

5 Numerical evidence and reproducibility

The interval tests use $(u, v, p, q) = (0.1, -0.1, 0.4, 0.4)$, $R = 0.25$, $M = 16, 64, 256$, and $r = -0.2, -10^{-12}, -10^{-24}, 0, 10^{-24}, 10^{-12}, 0.2$. All 21 complete kernels, evaluated at 100 digits, satisfy (7) after separately allowing for saved truncation tails. The merger uses `expm1` on the logarithmic ratio for $r \neq 0$ and the exact digamma series at zero.

Eleven points were compared with the unmerged representation at 140 digits and doubled cutoff. Their absolute differences range from about $3.14\,10^{-23}$ at low mass to about 10^{-75} at tiny nonzero offsets; the $M = 256, r = 0$ change is about $3.71\,10^{-98}$. The explicit allowance 10^{-65} concerns empirical rounding only. All values, including imaginary residuals, are retained.

For a direct local-integral diagnostic, take $f_k(r) = r^k(1 - (r/R)^2)^3$ on $[-R, R]$, zero outside, $k = 0, 1, 2$. These functions are C^2 and their target values are $(1, 0, 0)$. The full kernel gives:

M	$\int f_0 U\, dr$	$\int f_1 U\, dr$	$\int f_2 U\, dr$
64	0.4758802344	$7.45120\,10^{-5}$	0.0030419739
256	0.5580727442	$4.52997\,10^{-5}$	0.0034353921

The 24/48-point Gauss refinements use 144 complete complex kernel evaluations. The largest refinement change is about $3.11\,10^{-39}$, well above some propagated series tails (as small as 10^{-191}). Neither the tail nor the refinement difference certifies total quadrature/rounding accuracy. These finite masses are visibly far from the limiting delta action.

Separately, evaluating only the proved sine approximation for $L = 4, 8, 16, 32, 64, 128$ gives f_0 integrals approximately $0.46605, 0.68713, 0.94635, 1.00246, 0.999651, 1.000026$. These are not full-kernel computations at $M = e^L$. The distinction is recorded in the data schema.

An independent Decimal/Fraction replay passes 446 checks, including the elementary constants, exact companion polynomials, all saved quadrature nodes and input hashes. Numerical checks support the implementation; the uniform and distributional statements rest on the arguments above. All prior scientific files and frozen snapshots are retained. These are model calculations built from established formulas, not a claim of a new complete holographic dual.

References

[1] J. Liu et al., arXiv:1808.00612v3, Eqs. (3.35)–(3.38), the two-series crossing kernel.

[2] W. Groenevelt, arXiv:math/0306424v1, Propositions 4.4 and 4.7; removable terms and Dirichlet integrals.

[3] NIST DLMF, Section 5.7.6, digamma difference series.

[4] NIST DLMF, Section 5.11, log-Gamma, digamma and Gamma-ratio asymptotics. Checked 31 October 2026.